Data Science Program, George Washington University

Sample Midterm Exam

DATS 6101 (Introduction to Data Science)

Date: October 26, 2026

Instructor: Prof. Sushovan Majhi

Name: _____

Please read the following instructions carefully.

- This is a **sample paper** for practice only, at the real length, the real shape and the real difficulty, covering everything through Oct 19.
- It prints **30 problems. Answer any 25.** Each is worth **6 points**, so the paper is marked out of **150**, and if you answer more than 25 **your best 25 count**.
- You have **2 hours 30 minutes**, the full session.
- Most problems are multiple choice and **exactly one option is correct**. Mark it clearly (☑). The rest say what they want.
- The exam is **open-book** and **open-note**, but **printed material only**: the chapters covered, the labs whose solutions have been released, the homework covered, and your own notes.
- **Laptops**, **tablets**, **phones** and **watches** must be switched off and put away. No AI assistance and no communication of any kind.
- Nothing has to be computed that cannot be done on paper: every critical value a question needs is **printed in the question**. A correct number with no sentence around it earns part marks.
- All notation has its usual meaning. You may use standard results directly by stating them appropriately. **Read the instructions for individual questions carefully** before answering.

SAMPLE

Question	Points	Score
1	6	
2	6	
3	6	
4	6	
5	6	
6	6	
7	6	
8	6	
9	6	
10	6	
11	6	
12	6	
13	6	
14	6	
15	6	
16	6	
17	6	
18	6	
19	6	
20	6	
21	6	
22	6	
23	6	
24	6	
25	6	
26	6	
27	6	
28	6	
29	6	
30	6	
Total:	180	

SAMPLE

Part A Reading a claim

1. (6 points) A study reports mean satisfaction **7.2 (95% CI 6.9 to 7.5)**. Which statement is defensible?

   - □ 95% of respondents rated between 6.9 and 7.5
   - □ There is a 95% probability the true mean is between 6.9 and 7.5
   - □ If the study were repeated many times, about 95% of intervals built this way would contain the true mean
   - □ The next respondent will rate between 6.9 and 7.5 with 95% probability

2. (6 points) Exam scores have mean 72 and standard deviation 11, from $n = 400$ students, so the standard error is 0.55. A report says *"almost every student scored within about half a point of 72."* The error is

   - □ describing students with a number that describes the average of 400 of them
   - □ using $n = 400$ when the class is larger
   - □ using the standard error instead of the variance
   - □ forgetting to multiply by 1.96

3. (6 points) A comparison returns $p = 0.30$. The correct reading is

   - □ there is a 30% chance the null hypothesis is true
   - □ the two groups are equal
   - □ data at least this extreme are unremarkable if the null is true, which leaves both "no effect" and "too small a study" alive
   - □ the study has 30% power

4. (6 points) An app's rating is estimated from **40,000** reviews, all left voluntarily. The interval is extremely narrow. This tells you

   - □ the rating is accurate to within the interval
   - □ the sample is large enough for the CLT, so the estimate is unbiased
   - □ nothing, because ratings are ordinal
   - □ the rating is precise, and says nothing about whether reviewers resemble users

5. (6 points) You test 10,000 simulated datasets in which the null is *exactly* true, and count how many give $p < 0.05$. You should see about

   - □ 0
   - □ 50
   - □ 500
   - □ 9,500

SAMPLE

Part B Probability and random variables

6. (6 points) A condition affects **2%** of a population. A test detects it in 95% of those who have it and is negative in 90% of those who do not. Someone tests positive. The probability they have the condition is closest to

   - □ 2%
   - □ 16%
   - □ 50%
   - □ 95%

7. (6 points) X has variance 9. Then $\mathrm{Var}(3X - 2)$ is

   - □ 25
   - □ 27
   - □ 79
   - □ 81

8. (6 points) Which is a valid probability density on $[0, 3]$?

   - □ $f(x) = 2x/9$
   - □ $f(x) = 1/2$
   - □ $f(x) = 3 - x$
   - □ $f(x) = x/3$

9. (6 points) Of 500 site visits, 200 came from search and 300 from social; 40 of the search visits and 60 of the social visits converted. Converting and arriving from search are

   - □ dependent, because social produced more conversions
   - □ dependent, because the groups are different sizes
   - □ independent—the conversion rate is 20% in both groups
   - □ impossible to judge without a significance test

10. (6 points) Two fair dice are rolled. The events *"the first shows a six"* and *"the total is seven"* are

    - □ dependent, because both involve the first die
    - □ independent, because the total is equally likely to be 7 whatever the first die shows
    - □ dependent, because a six makes a large total more likely
    - □ mutually exclusive

SAMPLE

11. (6 points) **Short answer.** A delivery fee is \$0 with probability 0.5, \$5 with probability 0.3 and \$20 with probability 0.2. Compute $E[X]$ and $\mathrm{Var}(X)$, showing the working.

Part C Expectation, variance and the CLT

12. (6 points) An interval is currently ± 2.0 at 95% confidence. You want ± 1.0 at 90%, with σ unchanged. Then n must grow by a factor of about

$z_{0.975}$	1.960	$z_{0.95}$	1.645

- □ 2
- □ 2.8
- □ 3.4
- □ 4

13. (6 points) A clustered survey of **800** has a true standard error **twice** what $s/\sqrt{800}$ reports. Its effective sample size is about
    - □ 200
    - □ 400
    - □ 800
    - □ 1,600

14. (6 points) The central limit theorem says nothing about which of these?
    - □ the sampling distribution of $\bar{X}$ at large n
    - □ the mean of the sampling distribution of $\bar{X}$
    - □ the spread of the sampling distribution of $\bar{X}$
    - □ the sampling distribution of the sample *median*

SAMPLE

15. (6 points) The population is exponential, whose skewness is 2. You average samples of $n = 25$. The skewness of the sample means is closest to

    □ 0

    □ 0.4

    □ 1

    □ 2

16. (6 points) For a Cauchy population, the mean of 1,000 draws is no more precise than the mean of 10. This is because

    □ the draws are dependent

    □ the Cauchy is discrete

    □ 1,000 is still too small

    □ the Cauchy has no finite variance, so $\sigma/\sqrt{n}$ has nothing to describe

17. (6 points) Skewness of the sampling distribution of $\bar{X}$ falls like $1/\sqrt{n}$. To cut it to **one third**, you must multiply the sample size by

    □ 3

    □ 9

    □ 27

    □ 81

Part D Confidence intervals

18. (6 points) Drawing $n = 12$ from a strongly right-skewed population, a 95% t interval covers the true mean about 88% of the time. Which change most directly fixes the shortfall?

    □ a larger sample, so the sampling distribution of the mean is closer to Normal

    □ dividing by n rather than $n - 1$ when computing s

    □ `qnorm(0.975)` in place of `qt(0.975, 11)`

    □ a larger multiplier from the t table

SAMPLE

19. (6 points) **Short answer.** A sample of **100** tickets has $\bar{x} = 52$ and $s = 20$ minutes. Using $t_{0.975,99} = 1.984$, give the standard error and the 95% interval, with the working.

20. (6 points) A paper reports a mean of 12.4 with a 95% CI of 11.6 to 13.2 and a standard deviation of 4.0, but never gives the sample size. It is closest to

    □ 25

    □ 96

    □ 250

    □ 400

21. (6 points) Reporting a **90%** interval instead of a 95% one, everything else fixed, multiplies the width by about

    □ 0.84

    □ 0.90

    □ 0.95

    □ 1.19

22. (6 points) A trial converts **7 of 250**. `prop.test(7, 250, correct = FALSE)` returns an interval of 0.0136 to 0.0567 around $\hat{p} = 0.028$. It is **not** symmetric about $\hat{p}$ because

    □ of rounding in the endpoints

    □ `prop.test` applies a continuity correction

    □ the sample size is too small for symmetry

    □ a rate near zero has more room to be larger than smaller, and the Wilson interval reflects that

23. (6 points) Group A has mean 20.0 and group B 24.0, each with standard error 1.5. Using 1.96 and $SE_{diff} = \sqrt{1.5^2 + 1.5^2} = 2.12$, the 95% interval for the difference is

    □ 1.06 to 6.94, which excludes zero

    □ not computable, because the intervals overlap

    □ 0.94 to 7.06, which excludes zero

    □ -0.16 to 8.16, which includes zero

SAMPLE

24. (6 points) **Short answer.** An analyst's 95% interval for mean session length narrows pleasingly once they exclude every session longer than an hour. In one or two sentences, say what they have actually reported.

Part E Testing

25. (6 points) Thirty agents are timed on the same task before and after a training course. To ask whether the mean time changed, use

    - □ a two-sample (Welch) t-test
    - □ a one-sample t-test on the times recorded afterwards
    - □ a paired t-test
    - □ a chi-squared test

26. (6 points) You want to know whether complaint type (four categories) is related to region (three regions). The test is

    - □ a chi-squared test of independence on the 4×3 table
    - □ `prop.test` on the four complaint types
    - □ a two-sample t-test on each pair of regions
    - □ one-way ANOVA across the three regions

27. (6 points) An experiment on 2,000 sessions per arm returns $p = 0.002$ and a 95% confidence interval for the lift of **0.1% to 0.4%**. The right summary is

    - □ a 99.8% probability the new arm is better
    - □ a large, clearly valuable improvement
    - □ a difference that is hard to attribute to chance and too small to be worth much—significance and importance are different questions
    - □ an inconclusive result, since the interval is narrow

SAMPLE

28. (6 points) **Short answer.** A colleague reports *"no significant difference (*$p = 0.6$*,* $n = 18$ *per group)"* and proposes dropping the feature. In two sentences, say what is wrong with the inference and what you would ask for.

Part F Experiments

29. (6 points) Randomly assigning users to two versions buys you

    - □ groups alike in expectation on everything, measured or not, so a difference in outcome can be attributed to the version
    - □ freedom from the need for a confidence interval
    - □ identical groups
    - □ a larger effective sample size

30. (6 points) **Short answer.** In a randomized trial, 30% of the treatment arm stops using the product before the outcome is measured, against 10% of the control arm. In one or two sentences, say why the randomization no longer protects the comparison.

Thirty problems printed; your best twenty-five count.
A number with no working earns part marks; the sentence is what is being graded.

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